

HAT-P-32b

R-0256

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_230830 · created 2026-07-17T07:43:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460186.762 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.049
Transit depth (Rp/Rs)^2	1.7 +/- 1.29 [%]
Orbital Inclination (inc)	85.85 +/- 1.7
Airmass coefficient 1 (a1)	0.958 +/- 0.035
Airmass coefficient 2 (a2)	0.019 +/- 0.012
Scatter in the residuals of the lightcurve fit is	3.64 %
Best Comparison Star	#2 - [276, 297]
Optimal Aperture	3.58
Optimal Annulus	10.53
Transit Duration (day)	0.1215 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.131 ± 0.049 vs 0.1489 (deviation 0.31σ)
Duration fit vs published	0.1215 d vs 0.1304 d (deviation 0.95σ)
Mid-time O–C	-9.9 min
Depth SNR	2.3
Residual scatter	3.64 %

Light curve

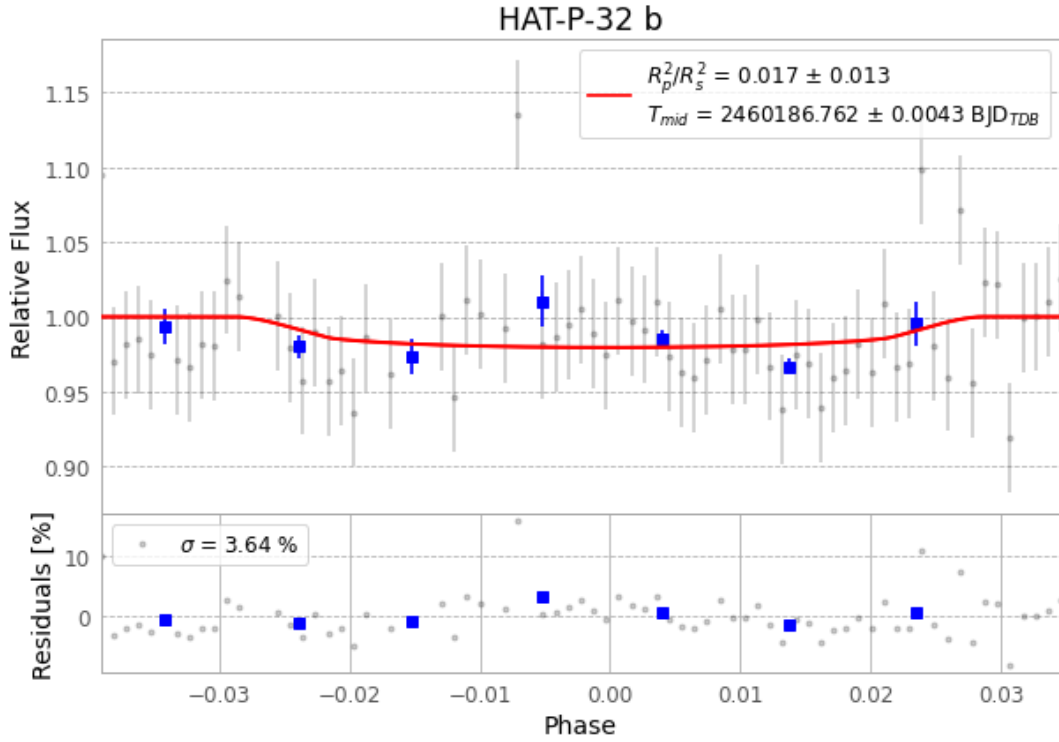


Figure 1 — FinalLightCurve_HAT-P-32 b_2023-08-29.png (SHA-256 5a4d4ad07d6d5e3c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [285, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8e4f128e1ad5b66f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 0eef8bc

Data provenance

77 FITS frames (51 MB), HATP-32230830041215.FITS → HATP-32230830080016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-230830.log (d452ea4ccb5a...), FinalLightCurve_HAT-P-32 b_2023-08-29.png (5a4d4ad07d6d...), FinalLightCurve_HAT-P-32 b_2023-08-29.csv (6dd39c271683...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)